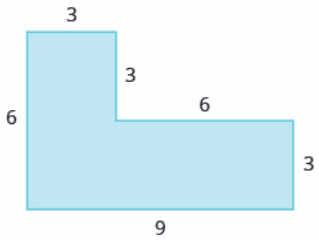
Writing Exercises

261. Describe two different ways to find the area of this figure, and then show your work to make sure both ways give the same area.



262. A circle has a diameter of 14 feet. Find the area of the circle (a) using 3.14 for π (b) using $\frac{22}{7}$ for π. (c) Which calculation to do prefer? Why?

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
use the properties of circles.			
find the area of irregular figures.			

ⓑ *After looking at the checklist, do you think you are well prepared for the next section? Why or why not?*

9.6 Solve Geometry Applications: Volume and Surface Area

Learning Objectives

By the end of this section, you will be able to:

- Find volume and surface area of rectangular solids
- Find volume and surface area of spheres
- Find volume and surface area of cylinders
- Find volume of cones

☑ **BE PREPARED** 9.16 Before you get started, take this readiness quiz.

Evaluate x^3 when $x = 5$.
If you missed this problem, review Example 2.15.

☑ **BE PREPARED** 9.17 Evaluate 2^x when $x = 5$.
If you missed this problem, review Example 2.16.

☑ **BE PREPARED** 9.18 Find the area of a circle with radius $\frac{7}{2}$.
If you missed this problem, review Example 5.39.

In this section, we will finish our study of geometry applications. We find the volume and surface area of some three-dimensional figures. Since we will be solving applications, we will once again show our Problem-Solving Strategy for Geometry Applications.

Problem Solving Strategy for Geometry Applications

Step 1. **Read** the problem and make sure you understand all the words and ideas. Draw the figure and label it with the given information.
Step 2. **Identify** what you are looking for.
Step 3. **Name** what you are looking for. Choose a variable to represent that quantity.
Step 4. **Translate** into an equation by writing the appropriate formula or model for the situation. Substitute in the given information.
Step 5. **Solve** the equation using good algebra techniques.
Step 6. **Check** the answer in the problem and make sure it makes sense.

Step 7. **Answer** the question with a complete sentence.

Find Volume and Surface Area of Rectangular Solids

A cheerleading coach is having the squad paint wooden crates with the school colors to stand on at the games. (See Figure 9.28). The amount of paint needed to cover the outside of each box is the **surface area**, a square measure of the total area of all the sides. The amount of space inside the crate is the volume, a cubic measure.

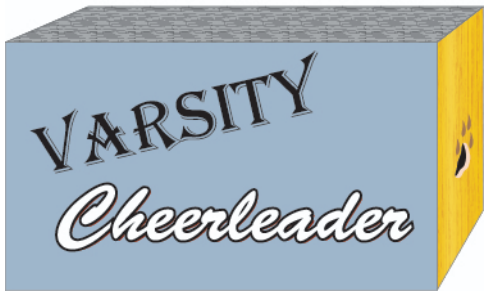


Figure 9.28 This wooden crate is in the shape of a rectangular solid.

Each crate is in the shape of a **rectangular solid**. Its dimensions are the length, width, and height. The rectangular solid shown in Figure 9.29 has length 4 units, width 2 units, and height 3 units. Can you tell how many cubic units there are altogether? Let's look layer by layer.

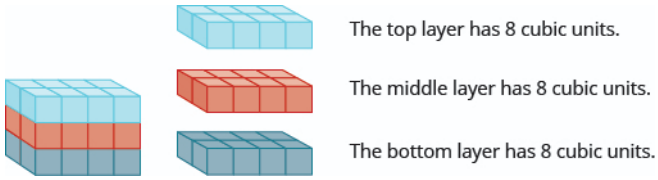


Figure 9.29 Breaking a rectangular solid into layers makes it easier to visualize the number of cubic units it contains. This 4 by 2 by 3 rectangular solid has 24 cubic units.

Altogether there are 24 cubic units. Notice that 24 is the $\text{length} \times \text{width} \times \text{height}$.

$$\underbrace{V}_{24} = \underbrace{L}_{4} \cdot \underbrace{W}_{2} \cdot \underbrace{H}_{3}$$

The volume, V, of any rectangular solid is the product of the length, width, and height.

$$V = LWH$$

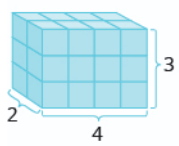
We could also write the formula for volume of a rectangular solid in terms of the area of the base. The area of the base, B, is equal to $\text{length} \times \text{width}$.

$$B = L \cdot W$$

We can substitute B for $L \cdot W$ in the volume formula to get another form of the volume formula.

$V = L \cdot W \cdot H$
$V = (L \cdot W) \cdot H$
$V = Bh$

We now have another version of the volume formula for rectangular solids. Let's see how this works with the $4 \times 2 \times 3$ rectangular solid we started with. See Figure 9.29.



$V = Bh$
$V = \text{Base} \times \text{height}$
$V = (4 \cdot 2) \times \text{height}$
$V = (4 \cdot 2) \times 3$
$V = 8 \times 3$
$V = 24$ cubic units

Figure 9.30

To find the *surface area* of a rectangular solid, think about finding the area of each of its faces. How many faces does the rectangular solid above have? You can see three of them.

$$A_{\text{front}} = L \times W \qquad A_{\text{side}} = L \times W \qquad A_{\text{top}} = L \times W$$
$$A_{\text{front}} = 4 \cdot 3 \qquad A_{\text{side}} = 2 \cdot 3 \qquad A_{\text{top}} = 4 \cdot 2$$
$$A_{\text{front}} = 12 \qquad A_{\text{side}} = 6 \qquad A_{\text{top}} = 8$$

Notice for each of the three faces you see, there is an identical opposite face that does not show.

$$S = (\text{front} + \text{back}) + (\text{left side} + \text{right side}) + (\text{top} + \text{bottom})$$
$$S = (2 \cdot \text{front}) + (2 \cdot \text{left side}) + (2 \cdot \text{top})$$
$$S = 2 \cdot 12 + 2 \cdot 6 + 2 \cdot 8$$
$$S = 24 + 12 + 16$$
$$S = 52 \text{ sq. units}$$

The surface area S of the rectangular solid shown in Figure 9.30 is 52 square units.

In general, to find the surface area of a rectangular solid, remember that each face is a rectangle, so its area is the product of its two dimensions, either length and width, length and height, or width and height (see Figure 9.31). Find the area of each face that you see and then multiply each area by two to account for the face on the opposite side.

$$S = 2LH + 2LW + 2WH$$

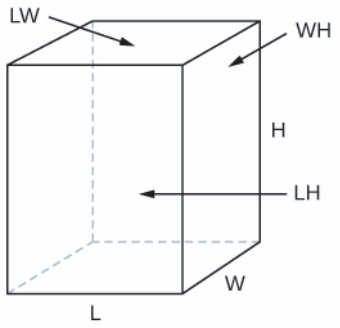
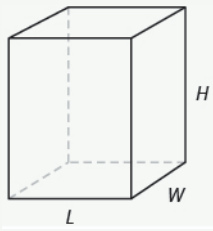


Figure 9.31 For each face of the rectangular solid facing you, there is another face on the opposite side. There are 6 faces in all.

Volume and Surface Area of a Rectangular Solid

For a rectangular solid with length L, width W, and height H :



Volume: $V = LWH$
Surface Area: $S = 2LH + 2LW + 2WH$

MANIPULATIVE MATHEMATICS

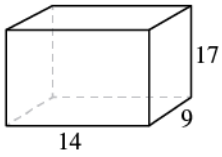
Doing the Manipulative Mathematics activity "Painted Cube" will help you develop a better understanding of volume and surface area.

EXAMPLE 9.47

For a rectangular solid with length 14 cm, height 17 cm, and width 9 cm, find the (a) volume and (b) surface area.

Solution

Step 1 is the same for both (a) and (b), so we will show it just once.

Step 1. **Read** the problem. Draw the figure and label it with the given information.	

(a)

Step 2. **Identify** what you are looking for.	the volume of the rectangular solid
Step 3. **Name.** Choose a variable to represent it.	Let V= volume
Step 4. **Translate.** Write the appropriate formula. Substitute.	 $V = LWH$ $V = 14 \cdot 9 \cdot 17$
Step 5. **Solve** the equation.	$V = 2{,}142$
Step 6. **Check** We leave it to you to check your calculations.	
Step 7. **Answer** the question.	The volume is $2{,}142$ cubic centimeters.

(b)

Step 2. **Identify** what you are looking for.	the surface area of the solid
Step 3. **Name.** Choose a variable to represent it.	Let S= surface area
Step 4. **Translate.** Write the appropriate formula. Substitute.	 $S = 2LH + 2LW + 2WH$ $S = 2(14 \cdot 17) + 2(14 \cdot 9) + 2(9 \cdot 17)$
Step 5. **Solve the equation.**	$S = 1{,}034$
Step 6. **Check:** Double-check with a calculator.	
Step 7. **Answer** the question.	The surface area is 1,034 square centimeters.

TRY IT 9.93 Find the (a) volume and (b) surface area of rectangular solid with the: length 8 feet, width 9 feet, and height 11 feet.

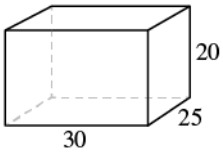
TRY IT 9.94 Find the (a) volume and (b) surface area of rectangular solid with the: length 15 feet, width 12 feet, and height 8 feet.

EXAMPLE 9.48

A rectangular crate has a length of 30 inches, width of 25 inches, and height of 20 inches. Find its (a) volume and (b) surface area.

Solution

Step 1 is the same for both (a) and (b), so we will show it just once.

Step 1. **Read** the problem. Draw the figure and label it with the given information.	

(a)

Step 2. **Identify** what you are looking for.	the volume of the crate
Step 3. **Name.** Choose a variable to represent it.	let V = volume
Step 4. **Translate.** Write the appropriate formula. Substitute.	 $V = LWH$ $V = 30 \cdot 25 \cdot 20$
Step 5. **Solve** the equation.	$V = 15{,}000$
Step 6. **Check:** Double check your math.	
Step 7. **Answer** the question.	The volume is 15,000 cubic inches.

(b)

Step 2. **Identify** what you are looking for.	the surface area of the crate
Step 3. **Name.** Choose a variable to represent it.	let S = surface area
Step 4. **Translate.** Write the appropriate formula. Substitute.	 $S = 2LH + 2LW + 2WH$ $S = 2(30 \cdot 20) + 2(30 \cdot 25) + 2(25 \cdot 20)$
Step 5. **Solve** the equation.	$S = 3{,}700$
Step 6. **Check:** Check it yourself!	
Step 7. **Answer** the question.	The surface area is 3,700 square inches.

TRY IT 9.95 A rectangular box has length 9 feet, width 4 feet, and height 6 feet. Find its (a) volume and (b) surface area.

TRY IT 9.96 A rectangular suitcase has length 22 inches, width 14 inches, and height 9 inches. Find its (a) volume and (b) surface area.

Volume and Surface Area of a Cube

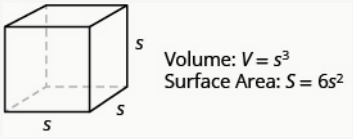
A **cube** is a rectangular solid whose length, width, and height are equal. See Volume and Surface Area of a Cube, below. Substituting, s for the length, width and height into the formulas for volume and surface area of a rectangular solid, we get:

$$\begin{aligned} V &= LWH \\ V &= s \cdot s \cdot s \\ V &= s^3 \end{aligned} \qquad \begin{aligned} S &= 2LH + 2LW + 2WH \\ S &= 2s \cdot s + 2s \cdot s + 2s \cdot s \\ S &= 2s^2 + 2s^2 + 2s^2 \\ S &= 6s^2 \end{aligned}$$

So for a cube, the formulas for volume and surface area are $V = s^3$ and $S = 6s^2$.

Volume and Surface Area of a Cube

For any cube with sides of length s,

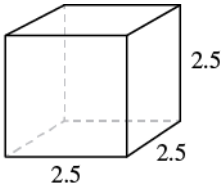


EXAMPLE 9.49

A cube is 2.5 inches on each side. Find its (a) volume and (b) surface area.

Solution

Step 1 is the same for both (a) and (b), so we will show it just once.

Step 1. **Read** the problem. Draw the figure and label it with the given information.	

(a)

Step 2. **Identify** what you are looking for.	the volume of the cube
Step 3. **Name.** Choose a variable to represent it.	let V = volume
Step 4. **Translate.** Write the appropriate formula.	$V = s^3$
Step 5. **Solve.** Substitute and solve.	$V = (2.5)^3$ $V = 15.625$

Step 6. **Check:** Check your work.	
Step 7. **Answer** the question.	The volume is 15.625 cubic inches.

ⓑ

Step 2. **Identify** what you are looking for.	the surface area of the cube
Step 3. **Name.** Choose a variable to represent it.	let S = surface area
Step 4. **Translate.** Write the appropriate formula.	$S = 6s^2$
Step 5. **Solve.** Substitute and solve.	$S = 6 \cdot (2.5)^2$ $S = 37.5$
Step 6. **Check:** The check is left to you.	
Step 7. **Answer** the question.	The surface area is 37.5 square inches.

TRY IT 9.97 For a cube with side 4.5 meters, find the ⓐ volume and ⓑ surface area of the cube.

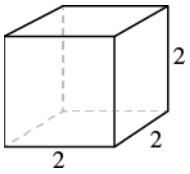
TRY IT 9.98 For a cube with side 7.3 yards, find the ⓐ volume and ⓑ surface area of the cube.

EXAMPLE 9.50

A notepad cube measures 2 inches on each side. Find its ⓐ volume and ⓑ surface area.

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.



ⓐ

Step 2. **Identify** what you are looking for.	the volume of the cube
Step 3. **Name.** Choose a variable to represent it.	let V = volume
Step 4. **Translate.** Write the appropriate formula.	$V = s^3$

Step 5. **Solve** the equation.	$V = 2^3$ $V = 8$
Step 6. **Check:** Check that you did the calculations correctly.	
Step 7. **Answer** the question.	The volume is 8 cubic inches.

ⓑ

Step 2. **Identify** what you are looking for.	the surface area of the cube
Step 3. **Name.** Choose a variable to represent it.	let S = surface area
Step 4. **Translate.** Write the appropriate formula.	$S = 6s^2$
Step 5. **Solve** the equation.	$S = 6 \cdot 2^2$ $S = 24$
Step 6. **Check:** The check is left to you.	
Step 7. **Answer** the question.	The surface area is 24 square inches.

TRY IT 9.99 A packing box is a cube measuring 4 feet on each side. Find its ⓐ volume and ⓑ surface area.

TRY IT 9.100 A wall is made up of cube-shaped bricks. Each cube is 16 inches on each side. Find the ⓐ volume and ⓑ surface area of each cube.

Find the Volume and Surface Area of Spheres

A **sphere** is the shape of a basketball, like a three-dimensional circle. Just like a circle, the size of a sphere is determined by its radius, which is the distance from the center of the sphere to any point on its surface. The formulas for the volume and surface area of a sphere are given below.

Showing where these formulas come from, like we did for a rectangular solid, is beyond the scope of this course. We will approximate π with 3.14.

Volume and Surface Area of a Sphere

For a sphere with radius r:



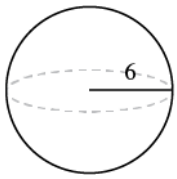
Volume: $V = \frac{4}{3}\pi r^3$
Surface Area: $S = 4\pi r^2$

EXAMPLE 9.51

A sphere has a radius 6 inches. Find its (a) volume and (b) surface area.

Solution

Step 1 is the same for both (a) and (b), so we will show it just once.

Step 1. **Read** the problem. Draw the figure and label it with the given information.	

(a)

Step 2. **Identify** what you are looking for.	the volume of the sphere
Step 3. **Name.** Choose a variable to represent it.	let V = volume
Step 4. **Translate.** Write the appropriate formula.	$V = \frac{4}{3}\pi r^3$
Step 5. **Solve.**	$V \approx \frac{4}{3}(3.14)6^3$ $V \approx 904.32$ cubic inches
Step 6. **Check:** Double-check your math on a calculator.	
Step 7. **Answer** the question.	The volume is approximately 904.32 cubic inches.

(b)

Step 2. **Identify** what you are looking for.	the surface area of the sphere
Step 3. **Name.** Choose a variable to represent it.	let S = surface area
Step 4. **Translate.** Write the appropriate formula.	$S = 4\pi r^2$
Step 5. **Solve.**	$S \approx 4(3.14)6^2$ $S \approx 452.16$
Step 6. **Check:** Double-check your math on a calculator	
Step 7. **Answer** the question.	The surface area is approximately 452.16 square inches.

TRY IT 9.101 Find the (a) volume and (b) surface area of a sphere with radius 3 centimeters.

TRY IT 9.102 Find the (a) volume and (b) surface area of each sphere with a radius of 1 foot

EXAMPLE 9.52

A globe of Earth is in the shape of a sphere with radius 14 inches. Find its (a) volume and (b) surface area. Round the answer to the nearest hundredth.

Solution

Step 1. **Read** the problem. Draw a figure with the given information and label it.



(a)

Step 2. **Identify** what you are looking for.	the volume of the sphere
Step 3. **Name.** Choose a variable to represent it.	let V = volume
Step 4. **Translate.** Write the appropriate formula. Substitute. (Use 3.14 for π)	$V = \frac{4}{3}\pi r^3$ $V \approx \frac{4}{3}(3.14)14^3$
Step 5. **Solve.**	$V \approx 11{,}488.21$
Step 6. **Check:** We leave it to you to check your calculations.	
Step 7. **Answer** the question.	The volume is approximately 11,488.21 cubic inches.

(b)

Step 2. **Identify** what you are looking for.	the surface area of the sphere
Step 3. **Name.** Choose a variable to represent it.	let S = surface area
Step 4. **Translate.** Write the appropriate formula. Substitute. (Use 3.14 for π)	$S = 4\pi r^2$ $S \approx 4(3.14)14^2$
Step 5. **Solve.**	$S \approx 2461.76$
Step 6. **Check:** We leave it to you to check your calculations.	
Step 7. **Answer** the question.	The surface area is approximately 2461.76 square inches.

TRY IT 9.103 A beach ball is in the shape of a sphere with radius of 9 inches. Find its (a) volume and (b)

surface area.

TRY IT 9.104 A Roman statue depicts Atlas holding a globe with radius of 1.5 feet. Find the (a) volume and (b) surface area of the globe.

Find the Volume and Surface Area of a Cylinder

If you have ever seen a can of soda, you know what a cylinder looks like. A **cylinder** is a solid figure with two parallel circles of the same size at the top and bottom. The top and bottom of a cylinder are called the bases. The height h of a cylinder is the distance between the two bases. For all the cylinders we will work with here, the sides and the height, h, will be perpendicular to the bases.

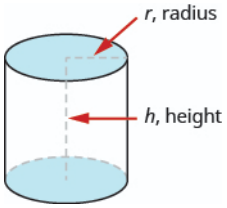


Figure 9.32 A cylinder has two circular bases of equal size. The height is the distance between the bases.

Rectangular solids and cylinders are somewhat similar because they both have two bases and a height. The formula for the volume of a rectangular solid, $V = Bh$, can also be used to find the volume of a cylinder.

For the rectangular solid, the area of the base, B, is the area of the rectangular base, length × width. For a cylinder, the area of the base, B, is the area of its circular base, πr^2. Figure 9.33 compares how the formula $V = Bh$ is used for rectangular solids and cylinders.

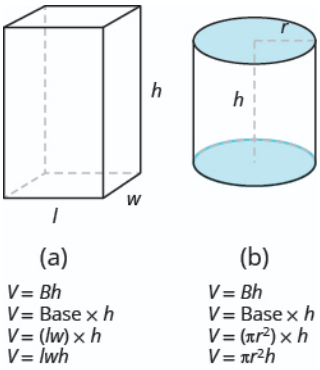


Figure 9.33 Seeing how a cylinder is similar to a rectangular solid may make it easier to understand the formula for the volume of a cylinder.

To understand the formula for the surface area of a cylinder, think of a can of vegetables. It has three surfaces: the top, the bottom, and the piece that forms the sides of the can. If you carefully cut the label off the side of the can and unroll it, you will see that it is a rectangle. See Figure 9.34.

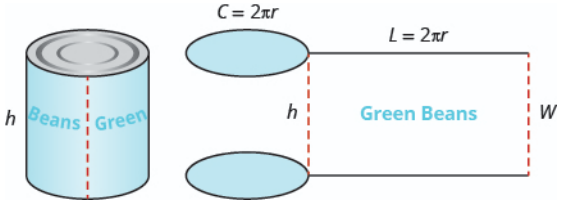
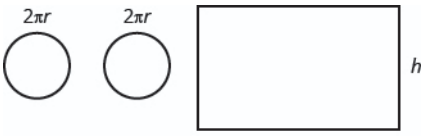


Figure 9.34 By cutting and unrolling the label of a can of vegetables, we can see that the surface of a cylinder is a rectangle. The length of the rectangle is the circumference of the cylinder's base, and the width is the height of the cylinder.

The distance around the edge of the can is the **circumference** of the cylinder's base it is also the length L of the rectangular label. The height of the cylinder is the width W of the rectangular label. So the area of the label can be represented as

$A = L \cdot W$
$A = 2\pi r \cdot h$

To find the total surface area of the cylinder, we add the areas of the two circles to the area of the rectangle.



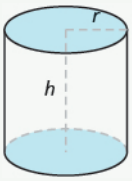
$S = A_{\text{top circle}} + A_{\text{bottom circle}} + A_{\text{rectangle}}$
$S = \underbrace{\pi r^2 + \pi r^2} + 2\pi r \cdot h$
$S = 2 \cdot \pi r^2 + 2\pi rh$
$S = 2\pi r^2 + 2\pi rh$

The surface area of a cylinder with radius r and height h, is

$$S = 2\pi r^2 + 2\pi rh$$

Volume and Surface Area of a Cylinder

For a cylinder with radius r and height h :



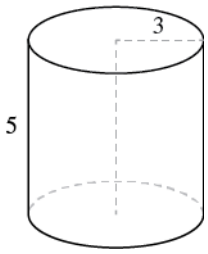
Volume: $V = \pi r^2 h$ or $V = Bh$
Surface Area: $S = 2\pi r^2 + 2\pi rh$

EXAMPLE 9.53

A cylinder has height 5 inches and radius 3 inches. Find the (a) volume and (b) surface area.

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.



(a)

Step 2. **Identify** what you are looking for.	the volume of the cylinder
Step 3. **Name.** Choose a variable to represent it.	let V = volume
Step 4. **Translate.** Write the appropriate formula. Substitute. (Use 3.14 for π)	 $V = \pi r^2 h$ $V \approx (3.14)3^2 \cdot 5$

Step 5. **Solve.**	$V \approx 141.3$
Step 6. **Check:** We leave it to you to check your calculations.	
Step 7. **Answer** the question.	The volume is approximately 141.3 cubic inches.

ⓑ

Step 2. **Identify** what you are looking for.	the surface area of the cylinder
Step 3. **Name.** Choose a variable to represent it.	let S = surface area
Step 4. **Translate.** Write the appropriate formula. Substitute. (Use 3.14 for π)	 $S = 2\pi r^2 + 2\pi rh$ $S \approx 2(3.14)3^2 + 2(3.14)(3)5$
Step 5. **Solve.**	$S \approx 150.72$
Step 6. **Check:** We leave it to you to check your calculations.	
Step 7. **Answer** the question.	The surface area is approximately 150.72 square inches.

TRY IT 9.105 Find the ⓐ volume and ⓑ surface area of the cylinder with radius 4 cm and height 7cm.

TRY IT 9.106 Find the ⓐ volume and ⓑ surface area of the cylinder with given radius 2 ft and height 8 ft.

EXAMPLE 9.54

Find the ⓐ volume and ⓑ surface area of a can of soda. The radius of the base is 4 centimeters and the height is 13 centimeters. Assume the can is shaped exactly like a cylinder.

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.	

ⓐ

Step 2. **Identify** what you are looking for.	the volume of the cylinder

Step 3. **Name.** Choose a variable to represent it.	let V = volume
Step 4. **Translate.** Write the appropriate formula. Substitute. (Use 3.14 for π)	 $V = \pi r^2 h$ $V \approx (3.14)4^2 \cdot 13$
Step 5. **Solve.**	$V \approx 653.12$
Step 6. **Check:** We leave it to you to check.	
Step 7. **Answer** the question.	The volume is approximately 653.12 cubic centimeters.

ⓑ

Step 2. **Identify** what you are looking for.	the surface area of the cylinder
Step 3. **Name.** Choose a variable to represent it.	let S = surface area
Step 4. **Translate.** Write the appropriate formula. Substitute. (Use 3.14 for π)	 $S = 2\pi r^2 + 2\pi rh$ $S \approx 2(3.14)4^2 + 2(3.14)(4)13$
Step 5. **Solve.**	$S \approx 427.04$
Step 6. **Check:** We leave it to you to check your calculations.	
Step 7. **Answer** the question.	The surface area is approximately 427.04 square centimeters.

TRY IT 9.107 Find the ⓐ volume and ⓑ surface area of a can of paint with radius 8 centimeters and height 19 centimeters. Assume the can is shaped exactly like a cylinder.

TRY IT 9.108 Find the ⓐ volume and ⓑ surface area of a cylindrical drum with radius 2.7 feet and height 4 feet. Assume the drum is shaped exactly like a cylinder.

Find the Volume of Cones

The first image that many of us have when we hear the word 'cone' is an ice cream cone. There are many other applications of cones (but most are not as tasty as ice cream cones). In this section, we will see how to find the volume of a cone.

In geometry, a **cone** is a solid figure with one circular base and a vertex. The height of a cone is the distance between its base and the vertex.The cones that we will look at in this section will always have the height perpendicular to the base. See Figure 9.35.

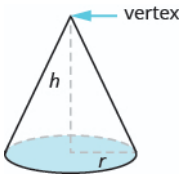


Figure 9.35 The height of a cone is the distance between its base and the vertex.

Earlier in this section, we saw that the volume of a cylinder is $V = \pi r^2 h$. We can think of a cone as part of a cylinder. Figure 9.36 shows a cone placed inside a cylinder with the same height and same base. If we compare the volume of the cone and the cylinder, we can see that the volume of the cone is less than that of the cylinder.

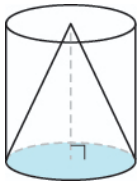


Figure 9.36 The volume of a cone is less than the volume of a cylinder with the same base and height.

In fact, the volume of a cone is exactly one-third of the volume of a cylinder with the same base and height. The volume of a cone is

$$V = \frac{1}{3}Bh$$

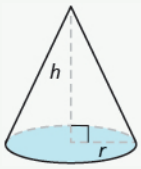
Since the base of a cone is a circle, we can substitute the formula of area of a circle, πr^2 , for B to get the formula for volume of a cone.

$$V = \frac{1}{3}\pi r^2 h$$

In this book, we will only find the volume of a cone, and not its surface area.

Volume of a Cone

For a cone with radius r and height h.

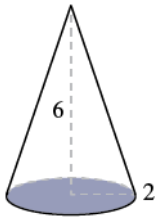


Volume: $V = \frac{1}{3}\pi r^2 h$

EXAMPLE 9.55

Find the volume of a cone with height 6 inches and radius of its base 2 inches.

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.	
Step 2. **Identify** what you are looking for.	the volume of the cone

Step 3. **Name.** Choose a variable to represent it.	let V = volume
Step 4. **Translate.** Write the appropriate formula. Substitute. (Use 3.14 for π)	$V = \frac{1}{3} \quad \pi \quad r^2 \quad h$ $V \approx \frac{1}{3} \quad 3.14 \quad (2)^2 \quad (6)$
Step 5. **Solve.**	$V \approx 25.12$
Step 6. **Check:** We leave it to you to check your calculations.	
Step 7. **Answer** the question.	The volume is approximately 25.12 cubic inches.

TRY IT 9.109 Find the volume of a cone with height 7 inches and radius 3 inches

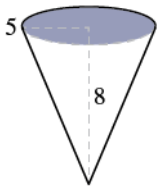
TRY IT 9.110 Find the volume of a cone with height 9 centimeters and radius 5 centimeters

EXAMPLE 9.56

Marty's favorite gastro pub serves french fries in a paper wrap shaped like a cone. What is the volume of a conic wrap that is 8 inches tall and 5 inches in diameter? Round the answer to the nearest hundredth.

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information. Notice here that the base is the circle at the top of the cone.



Step 2. **Identify** what you are looking for.	the volume of the cone
Step 3. **Name.** Choose a variable to represent it.	let V = volume
Step 4. **Translate.** Write the appropriate formula. Substitute. (Use 3.14 for π, and notice that we were given the distance across the circle, which is its diameter. The radius is 2.5 inches.)	$V = \frac{1}{3} \quad \pi \quad r^2 \quad h$ $V \approx \frac{1}{3} \quad 3.14 \quad (2.5)^2 \quad (8)$
Step 5. **Solve.**	$V \approx 52.33$
Step 6. **Check:** We leave it to you to check your calculations.	
Step 7. **Answer** the question.	The volume of the wrap is approximately 52.33 cubic inches.

TRY IT 9.111 How many cubic inches of candy will fit in a cone-shaped piñata that is 18 inches long and 12

inches across its base? Round the answer to the nearest hundredth.

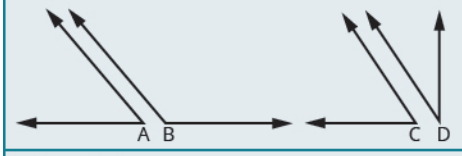
TRY IT 9.112 What is the volume of a cone-shaped party hat that is 10 inches tall and 7 inches across at the base? Round the answer to the nearest hundredth.

Summary of Geometry Formulas

The following charts summarize all of the formulas covered in this chapter.

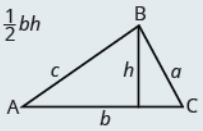
Supplementary and Complementary Angles

$m\angle A + m\angle B = 180°$ for supplementary angles A and B
$m\angle C + m\angle D = 90°$ for complementary angles C and D



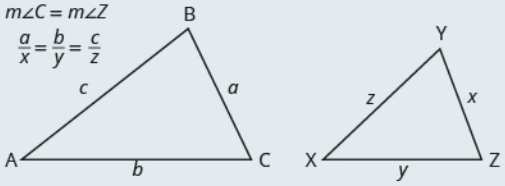
Triangle

For ΔABC, angle measures.
$m\angle A + m\angle B + m\angle C = 180°$
Perimeter. $P = a + b + c$
$A = \frac{1}{2}bh$



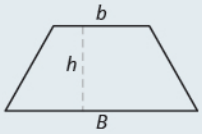
Similar Triangles

$m\angle A = m\angle X$
$m\angle B = m\angle Y$
$m\angle C = m\angle Z$
$\frac{a}{x} = \frac{b}{y} = \frac{c}{z}$



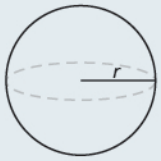
Trapezoid

Area. $A = \frac{1}{2}h(b + B)$



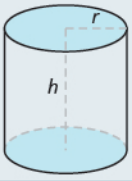
Sphere

Volume: $V = \frac{4}{3}\pi r^3$
Surface Area: $S = 4\pi r^2$



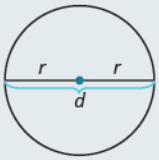
Cylinder

Volume: $V = \pi r^2 h$ or $V = Bh$
Surface Area: $S = 2\pi r^2 + 2\pi rh$

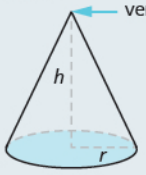


Circle

Circumference: $C = 2\pi r$ or $C = \pi d$
Area: $A = \pi r^2$



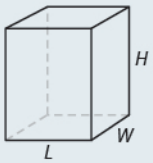
Cone



$V = \frac{1}{3}\pi r^2 h$

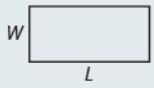
Rectangular Solid

Volume: $V = LWH$
Surface Area: $S = 2LH + 2LW + 2WH$



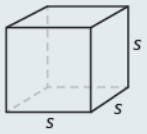
Rectangle

Perimeter: $P = 2L + 2W$
Area: $A = LW$



Cube

Volume: $V = s^3$
Surface Area: $S = 6s^2$



MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Volume of a Cone (https://openstaxcollege.org/l/24volcone)

SECTION 9.6 EXERCISES

Practice Makes Perfect

Find Volume and Surface Area of Rectangular Solids

In the following exercises, find ⓐ the volume and ⓑ the surface area of the rectangular solid with the given dimensions.

263. length 2 meters, width 1.5 meters, height 3 meters

264. length 5 feet, width 8 feet, height 2.5 feet

265. length 3.5 yards, width 2.1 yards, height 2.4 yards

266. length 8.8 centimeters, width 6.5 centimeters, height 4.2 centimeters

In the following exercises, solve.

267. **Moving van** A rectangular moving van has length 16 feet, width 8 feet, and height 8 feet. Find its (a) volume and (b) surface area.

268. **Gift box** A rectangular gift box has length 26 inches, width 16 inches, and height 4 inches. Find its (a) volume and (b) surface area.

269. **Carton** A rectangular carton has length 21.3 cm, width 24.2 cm, and height 6.5 cm. Find its (a) volume and (b) surface area.

270. **Shipping container** A rectangular shipping container has length 22.8 feet, width 8.5 feet, and height 8.2 feet. Find its (a) volume and (b) surface area.

In the following exercises, find ⓐ the volume and ⓑ the surface area of the cube with the given side length.

271. 5 centimeters

272. 6 inches

273. 10.4 feet

274. 12.5 meters

In the following exercises, solve.

275. **Science center** Each side of the cube at the Discovery Science Center in Santa Ana is 64 feet long. Find its (a) volume and (b) surface area.

276. **Museum** A cube-shaped museum has sides 45 meters long. Find its (a) volume and (b) surface area.

277. **Base of statue** The base of a statue is a cube with sides 2.8 meters long. Find its (a) volume and (b) surface area.

278. **Tissue box** A box of tissues is a cube with sides 4.5 inches long. Find its (a) volume and (b) surface area.

Find the Volume and Surface Area of Spheres

In the following exercises, find ⓐ the volume and ⓑ the surface area of the sphere with the given radius. Round answers to the nearest hundredth.

279. 3 centimeters

280. 9 inches

281. 7.5 feet

282. 2.1 yards

In the following exercises, solve. Round answers to the nearest hundredth.

283. **Exercise ball** An exercise ball has a radius of 15 inches. Find its (a) volume and (b) surface area.

284. **Balloon ride** The Great Park Balloon is a big orange sphere with a radius of 36 feet . Find its (a) volume and (b) surface area.

285. **Golf ball** A golf ball has a radius of 4.5 centimeters. Find its (a) volume and (b) surface area.

286. **Baseball** A baseball has a radius of 2.9 inches. Find its (a) volume and (b) surface area.

Find the Volume and Surface Area of a Cylinder

In the following exercises, find (a) the volume and (b) the surface area of the cylinder with the given radius and height. Round answers to the nearest hundredth.

287. radius 3 feet, height 9 feet

288. radius 5 centimeters, height 15 centimeters

289. radius 1.5 meters, height 4.2 meters

290. radius 1.3 yards, height 2.8 yards

In the following exercises, solve. Round answers to the nearest hundredth.

291. **Coffee can** A can of coffee has a radius of 5 cm and a height of 13 cm. Find its (a) volume and (b) surface area.

292. **Snack pack** A snack pack of cookies is shaped like a cylinder with radius 4 cm and height 3 cm. Find its (a) volume and (b) surface area.

293. **Barber shop pole** A cylindrical barber shop pole has a diameter of 6 inches and height of 24 inches. Find its (a) volume and (b) surface area.

294. **Architecture** A cylindrical column has a diameter of 8 feet and a height of 28 feet. Find its (a) volume and (b) surface area.

Find the Volume of Cones

In the following exercises, find the volume of the cone with the given dimensions. Round answers to the nearest hundredth.

295. height 9 feet and radius 2 feet

296. height 8 inches and radius 6 inches

297. height 12.4 centimeters and radius 5 cm

298. height 15.2 meters and radius 4 meters

In the following exercises, solve. Round answers to the nearest hundredth.

299. **Teepee** What is the volume of a cone-shaped teepee tent that is 10 feet tall and 10 feet across at the base?

300. **Popcorn cup** What is the volume of a cone-shaped popcorn cup that is 8 inches tall and 6 inches across at the base?

301. **Silo** What is the volume of a cone-shaped silo that is 50 feet tall and 70 feet across at the base?